An Extended Double Threshold-Based Detection Method for Low-SNR Maritime Scene Targets in Range-Compressed Spaceborne SAR Data

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***Abstract*—Spaceborne synthetic aperture radar (SAR) target detection in maritime scenes is a critical task for ocean remote sensing. Conventional detection methods typically operate on SAR imagery and rely on full-aperture azimuth focusing, which entails high computational costs and limits processing timeliness. In contrast, detection in the range-compressed domain before imaging offers significant efficiency advantages. However, for weak maritime targets, the range-compressed echoes are easily obscured by strong sea clutter and suffer from range migration. To address these issues, this article proposes a range-compressed domain detection method that cascades correct-before-detect (CBD) with an extended double threshold constant false alarm rate (EDT-CFAR) detector. First, the CBD method is developed by incorporating the fast Radon transform based on peak regression (PR-FRT) to efficiently correct range migration. Furthermore, the EDT-CFAR detector is designed under the generalized gamma distribution (GΓD) sea clutter model. Specifically, the first threshold executes cell-level decisions based on GΓD parameter estimation to screen candidate target cells while partially suppressing sea clutter. The second threshold constructs test statistics from the candidate cells and implements an adaptive decision mechanism under the CFAR criterion, thereby enabling weak target detection. Both simulations and experiments on Gaofen-3 (GF-3) measured data demonstrate that the proposed method enhances detection performance under low signal-to-noise ratio (SNR) conditions, providing a feasible approach for spaceborne SAR maritime scene target detection.**

***Index Terms*—Constant false alarm rate (CFAR), maritime scene target detection, range-compressed domain, spaceborne synthetic aperture radar (SAR), weak targets.**

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I. Introduction

SPACEBORNE synthetic aperture radar (SAR), benefiting from its all-day and all-weather observation capability, has been widely applied in marine monitoring and maritime situational awareness [1], [2]. For maritime scene target detection, current research primarily concentrates on medium to large-scale ships with relatively high signal-to-noise ratios (SNRs) and typically employs SAR images as input for target detection and recognition. Existing image domain detection methods generally fall into two categories. The first involves statistical model-based methods, which construct test statistics based on sea clutter models and determine thresholds for a specified false alarm probability via analytical derivation, numerical solutions, or statistical simulations [3], [4], [5], [6], [7]. The second comprises deep learning-based methods. These methods achieve target detection through feature learning and frequently incorporate mechanisms such as multiscale feature fusion and attention to enhance detection performance under complex sea states [8], [9], [10], [11], [12], [13]. The aforementioned studies have established a systematic technical foundation for the detection of medium and large-scale ships. However, for weak targets such as small-scale ships and aircraft in maritime scenes, the echo energy is relatively low due to the long observation range between the spaceborne SAR and the targets, as well as their generally low radar cross sections (RCSs). Furthermore, these targets often exhibit non-negligible motion characteristics. For instance, small-scale ships are susceptible to attitude perturbations induced by sea waves, while aircraft typically move at relatively high speeds. These motions can lead to pronounced defocusing in SAR images. Consequently, the target energy is dispersed and easily buried in sea clutter, rendering conventional image domain detection methods inadequate for weak target detection in maritime scenes. In addition, the limited computational efficiency of image domain detection has become increasingly evident in wide-area maritime target detection.

With the continued evolution of high-resolution wide-swath (HRWS) SAR systems [14], [15], [16], [17], the significantly enhanced wide-area observation capability has inevitably led to an increase in single scene data volume and backend processing complexity [18], [19], [20]. Conventional image domain detection methods typically follow a processing paradigm of full scene imaging followed by target detection,

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which overlooks the sparse target distribution in maritime scenes [21], [22]. This paradigm entails significant computational costs associated with azimuth focusing over an extensive sea clutter background, thereby severely limiting the timeliness of target detection in wide-area maritime scenes. In view of this, a more efficient strategy is to perform detection in the range-compressed domain before imaging. Specifically, after range compression, target detection is conducted directly by exploiting the statistical and structural features of the range-compressed data, with subsequent fine imaging performed only for the candidate regions. Nevertheless, this strategy presents its own challenges, including the relatively low SNR of range-compressed echoes and the energy spreading across range cells caused by range migration.

Range-compressed domain maritime target detection has drawn growing attention in recent years. In [23], two ship detection networks based on the faster region-based convolutional neural network (R-CNN) were developed in the time domain and the Doppler domain, respectively, enabling high recall performance in complex multitarget scenes. The Incept-text convolutional neural network (TextCNN) in [24] constructs a 1-D range profile signal from range-compressed data and introduces a multiscale feature fusion mechanism, achieving fast detection of the range cells containing ship targets. The YOLO-range compressed network in [25] incorporates an amplitude gradient guided feature extraction module to strengthen structural feature extraction, alleviating performance degradation caused by insufficient exploitation of critical characteristics in the range-compressed domain. Although these deep learning-based methods improve detection performance to some extent, their robustness and cross-scene generalization are often constrained by scarce labeled samples and limited discriminative echo features in the range-compressed domain. Furthermore, enforcing constant false alarm rate (CFAR) control remains challenging for these methods. A CFAR ship detection method was proposed in [26] and [27], which conducts target predetection in the time domain to remove bright peaks, followed by clutter normalization and statistical model parameter estimation in the Doppler domain to calculate CFAR thresholds. However, this method typically requires the target to possess a sufficient line-of-sight velocity component, and the azimuth Fourier transform (FT) still imposes substantial computational costs. The range-compressed-generalized gamma distribution-CFAR (RC-GΓD-CFAR) method in [28] derived analytical approximations for the parameters of the sea clutter model and designed a CFAR ship detector for the range-compressed domain, achieving effective ship detection under different sea states. Although capable of fast CFAR detection, this method fails to fully exploit the target energy distribution characteristics in the range-compressed domain. Notably, existing methods are predominantly developed for ships under relatively high SNR conditions, with limited emphasis on enhancing the detection performance of weak maritime targets and insufficient consideration of range migration correction.

To address these limitations, this article proposes a detection method for spaceborne SAR range-compressed data by cascading correct-before-detect (CBD) with an extended double threshold CFAR (EDT-CFAR) detector. First, a fast Radon transform based on peak regression (PR-FRT) is proposed for efficient range velocity estimation, which underpins the CBD processing for range migration correction. Building on this, inspired by the hierarchical screening and cascaded decision strategy in [29], the EDT-CFAR detector is constructed based on the generalized gamma distribution (GΓD) sea clutter model. Specifically, the first threshold derives an analytical decision threshold via approximate estimation of GΓD parameters and performs cell-by-cell decisions on the range-compressed data to extract target cells while suppressing a portion of sea clutter. The second threshold constructs test statistics from the candidate cells that pass the first threshold and theoretically derives the adaptive decision threshold under the CFAR criterion, thereby effectively suppressing false alarms caused by sea spikes and enabling the detection of weak targets. The proposed integrated method significantly alleviates the performance limitations of existing range-compressed domain detection methods imposed by target motion and low SNR, and achieves effective CFAR detection of weak targets in maritime scenes.

The main contributions of this article are summarized as follows.

1) A CBD method based on PR-FRT is proposed, which exploits the local intrinsic properties in the Radon transform (RT) domain to formulate a guided mechanism, thereby achieving efficient range migration correction.
2) An EDT-CFAR detector is designed for range-compressed data. Under the CFAR criterion, the double thresholds are analytically derived based on the GΓD sea clutter model, reducing the SNR requirement by approximately 2.5 dB compared with existing methods.
3) Gaofen-3 (GF-3) measured data validate that the integrated detection method can achieve effective detection of ships and aircraft over the sea background, extending the scope of maritime scene target detection to include aerial targets.

The remainder of this article is organized as follows. Section II analyzes the spaceborne SAR range-compressed signal model. Section III presents the CBD method. Section IV proposes the EDT-CFAR detector. Section V provides simulation analysis and experimental validation of the proposed method. Finally, Section VI concludes this article.

II. Preliminaries

This section analyzes the spaceborne SAR signal model in the range-compressed domain. The geometric relationship between the SAR platform and the maritime scene target is illustrated in Fig. 1. The SAR platform travels along the azimuth direction with a constant velocity of v_{st}. To highlight the impact of target motion on the range-compressed signal, an aircraft over a sea background is taken as an example for the derivation. Let v_a and v_r denote the azimuth and range velocities of the aircraft, respectively. Furthermore, let R_0 represent the near range between the SAR platform and the aircraft, and $R(\eta)$ denote the instantaneous slant range at azimuth time η.

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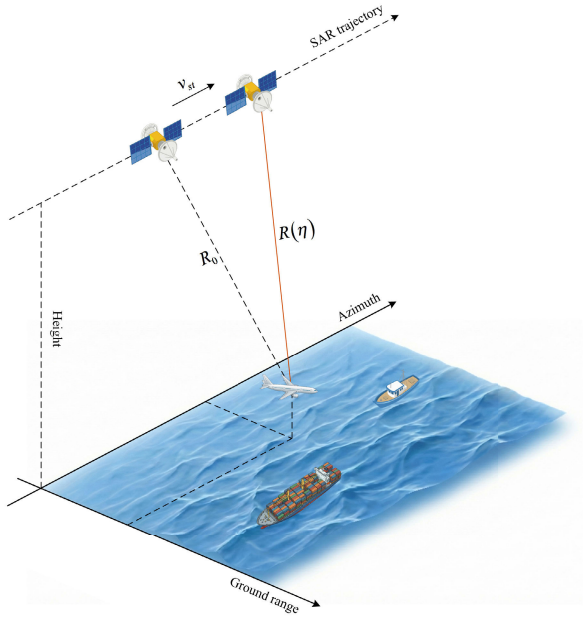


Fig. 1. Geometry between the spaceborne SAR platform and the maritime scene target.

The transmitted linear frequency-modulated (LFM) signal is given by

$$s(\tau) = w_r(\tau)\exp\left(j2\pi f_0\tau + j\pi k_r\tau^2\right) \tag{1}$$

where $w_r(\tau)$ is defined as

$$w_r(\tau) = \text{rect}\left(\frac{\tau}{T_p}\right) = \begin{cases} 1, & |\tau| < T_p/2 \\ 0, & |\tau| > T_p/2. \end{cases} \tag{2}$$

where τ is the fast time, T_p is the pulse duration, f_0 is the carrier frequency, and k_r is the frequency-modulated rate of the transmitted signal.

The instantaneous slant range between the SAR platform and the aircraft is

$$R(\eta) = \sqrt{(v_{st}\eta - v_a\eta)^2 + (R_0 + v_r\eta)^2}. \tag{3}$$

Performing a second-order Taylor expansion of $R(\eta)$ at $\eta = 0$ yields

$$R(\eta) = R_0 + v_r\eta + \frac{(v_{st} - v_a)^2}{2R_0}\eta^2. \tag{4}$$

Consequently, the received echo can be expressed as

$$s_r(\tau,\eta) = \sigma_0 s\left(\tau - \frac{2R(\eta)}{c}\right) \tag{5}$$

where c is the speed of light, η is the azimuth time, and σ_0 is the magnitude of backscatter coefficient.

The demodulated baseband echo signal can be expressed as

$$s_0(\tau,\eta) = \sigma_0 w_r\left(\tau - \frac{2R(\eta)}{c}\right)\exp\left(j\pi k_r\left(\tau - \frac{2R(\eta)}{c}\right)^2\right) \times \exp\left(-j\frac{4\pi}{\lambda}R(\eta)\right) \tag{6}$$

where λ is the wavelength of the transmitted signal.

After range compression, the time domain signal can be expressed as

$$s_{rc}(\tau,\eta) = G_{rc}\sigma_0\,\text{sinc}\left(B\left(\tau - \frac{2R(\eta)}{c}\right)\right) \times \exp\left(-j\frac{4\pi}{\lambda}R(\eta)\right) \tag{7}$$

where B is the signal bandwidth, and G_{rc} is the gain of range compression, with $G_{rc} = |k_r|{T_p}^2$.

It follows from (7) that the envelope peak of the range-compressed signal is distributed along $\tau = (2/c)(R_0 + v_r\eta + (((v_{st} - v_a)^2)/2R_0)\eta^2)$. During the exposure time, the envelope peak migrates through range cells, resulting in range migration. Specifically, the first-order term of η corresponds to range walk, while the second-order term corresponds to range curvature. Given that the spaceborne SAR platform velocity v_{st} is much higher than the aircraft azimuth velocity v_a, the range curvature is primarily caused by the high-speed platform motion.

III. Correct-Before-Detect

This section presents a CBD strategy for range migration correction. First, the dominant term of range curvature is compensated based on the prior platform velocity information. Subsequently, PR-FRT is proposed to efficiently estimate the target range velocity, thereby enabling range walk correction.

A. Fast RT Based on Peak Regression

The following filter is employed to correct the range curvature caused by the platform velocity:

$$H_{rcc}(f_\tau,\eta) = \exp\left(j\frac{2\pi}{c}f_\tau\frac{{v_{st}}^2}{R_0}\eta^2\right). \tag{8}$$

By performing range FT on the range-compressed signal and multiplying by $H_{rcc}(f_\tau,\eta)$, followed by range inverse FT (IFT), the time domain signal after range curvature precorrection can be expressed as:

$$s_{rcc}(\tau,\eta) = G_{rc}\sigma_0\exp\left(-j\frac{4\pi}{\lambda}R(\eta)\right) \times \text{sinc}\left(B\left(\tau - \frac{2}{c}\left(R_0 + v_r\eta + \frac{v_a^2 - 2v_{st}v_a}{2R_0}\eta^2\right)\right)\right). \tag{9}$$

After range curvature precorrection, the residual range migration can be approximated as a line structure, as shown in Fig. 2(a). RT is capable of projecting a line structure in a 2-D plane and estimating its location and angle, which can be used to correct the range walk. RT of the range curvature precorrected signal is defined as follows [30]:

$$R(\rho,\theta) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}|s_{rcc}(x,y)|\,\delta(\rho - x\cos\theta - y\sin\theta)\,dxdy \tag{10}$$

where ρ is the range from the line to the origin, θ is the angle between the normal direction of the line and the x-axis, and $\delta(\cdot)$ denotes the impulse function. By integrating the 2-D

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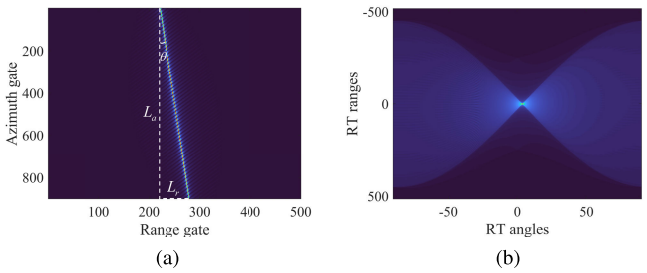


Fig. 2. Illustration of RT. (a) Geometric interpretation of range walk. (b) RT search results.

signal $|s_{rcc}(x, y)|$ along lines satisfying $x\cos\theta + y\sin\theta = \rho$, the projection function $R(\rho, \theta)$ is obtained. The parameters $(\hat{\rho}, \hat{\theta})$ corresponding to the maximum value of $R(\rho, \theta)$ determine the trajectory of the range walk, as shown in Fig. 2(b).

Since the traditional RT requires exhaustive search in the parameter space, resulting in high computational costs, this article proposes a PR-FRT method. First, RT of the signal $s_{rcc}(\tilde{\tau}, \tilde{\eta})$ within the detection window is expressed as

$$R(\rho,\theta) = \iint G_{rc}\sigma_0 \left|\operatorname{sinc}\left(B\left(\tilde{\tau} - \tau_0 - \frac{2v_r}{c}\tilde{\eta}\right)\right)\right| \times \delta(\rho - \tilde{\tau}\cos\theta - \tilde{\eta}\sin\theta)\,\mathrm{d}\tilde{\tau}\mathrm{d}\tilde{\eta} \tag{11}$$

where (τ_c, η_c) denotes the center of the detection window in the range-compressed data, and $(\tilde{\tau}, \tilde{\eta})$ denotes the corresponding local coordinates within the detection window, defined as $\tilde{\tau} = \tau - \tau_c$ and $\tilde{\eta} = \eta - \eta_c$. In addition, $\tau_0 = 2R_0/c + 2v_r\eta_c/c - \tau_c$.

To explicitly express the geometric constraints of the impulse function, a rotational coordinate transform consistent with the Radon line is introduced as

$$x' = \tilde{\tau}\cos\theta + \tilde{\eta}\sin\theta \tag{12}$$

$$y' = -\tilde{\tau}\sin\theta + \tilde{\eta}\cos\theta. \tag{13}$$

After the coordinate transform, RT becomes

$$R(\rho,\theta) = G_{rc}\sigma_0 \int \left|\operatorname{sinc}\left(B\left(\varphi(\rho,\theta) - y'\beta(\theta)\right)\right)\right| dy' \tag{14}$$

where

$$\varphi(\rho,\theta) = \rho\alpha(\theta) - \tau_0 \tag{15}$$

$$\alpha(\theta) = \cos\theta - \frac{2v_r}{c}\sin\theta \tag{16}$$

$$\beta(\theta) = \sin\theta + \frac{2v_r}{c}\cos\theta. \tag{17}$$

For each angle θ, the peak response of RT corresponding to that angle is selected as the objective function

$$f(\theta) = \max_{\rho}\{R(\rho,\theta)\}. \tag{18}$$

According to (14), it can be observed that for a fixed angle θ, the parameter ρ only induces a lateral translation $\varphi(\rho, \theta)$ of the envelope, whereas the mismatch stretching along the integration direction is controlled by $\beta(\theta)$. Under the condition $\alpha(\theta) \neq 0$ and neglecting finite-window boundary effects, $R(\rho, \theta)$ is approximately maximized at $\varphi(\rho, \theta) = 0$, i.e., $\hat{\rho}(\theta) = \tau_0/\alpha(\theta)$. Substituting this into (18) yields

$$f(\theta) \approx G_{rc}\sigma_0 \int_{-l/2}^{l/2} \left|\operatorname{sinc}\left(B\left(y'\beta(\theta)\right)\right)\right| dy' \tag{19}$$

where l denotes the equivalent integration length.

Therefore, when $\beta(\theta) = 0$, the integration kernel is decoupled from y', and the objective function $f(\theta)$ attains its global maximum at the optimal angle $\hat{\theta}$. To address the nondifferentiability of the $|\operatorname{sinc}(\cdot)|$ function at its zero crossings, a smoothing approximation [31] is adopted. By introducing a smoothing factor ε_s, (19) is modified to

$$f(\theta) \approx G_{rc}\sigma_0 \int_{-l/2}^{l/2} \sqrt{\left(\operatorname{sinc}\left(B\left(y'\beta(\theta)\right)\right)\right)^2 + {\varepsilon_s}^2}\, dy'. \tag{20}$$

Furthermore, the second-order Taylor expansion of $f(\theta)$ at $\theta = \hat{\theta}$ yields

$$f(\theta) \approx C_0 + C_1\left(\theta - \hat{\theta}\right)^2 + o\left(\left(\theta - \hat{\theta}\right)^2\right) \tag{21}$$

where

$$C_0 = G_{rc}\sigma_0 l\sqrt{1 + {\varepsilon_s}^2} \tag{22}$$

$$C_1 = -\frac{\pi^2}{72\sqrt{1 + {\varepsilon_s}^2}} G_{rc}\sigma_0 B^2 l^3 \left(\alpha\left(\hat{\theta}\right)\right)^2. \tag{23}$$

Based on the physical interpretation of the parameters, it can be inferred that $C_0 > 0$ and $C_1 < 0$. The analysis indicates that $f(\theta)$ exhibits a unimodal property in the neighborhood of $\hat{\theta}$, with its local geometry characterized by a quadratic curve centered at the vertex $\hat{\theta}$. Meanwhile, according to the geometric property of RT, in regions far from $\hat{\theta}$, a pronounced angular mismatch exists between the integration path and the line. Consequently, the signal energy cannot be accumulated effectively, resulting in a rapid decay of $f(\theta)$. Therefore, the angle estimation problem can be transformed into a peak localization problem based on a local quadratic model.

Instead of relying on the exhaustive search in the entire parameter space, the proposed method leverages the aforementioned analytical properties to construct a scale-contraction iterative mechanism. By adjusting the center and scale of the search window, the optimal solution of the objective function is approached iteratively, thereby enabling efficient estimation of the optimal angle $\hat{\theta}$.

Specifically, let $\theta_c^{(k)}$ and $R^{(k)}$ denote the estimation center and the search radius at the kth iteration, respectively. A uniformly discrete observation set $\boldsymbol{\Theta}_{(k)} = [\theta_1^{(k)}, \theta_2^{(k)}, \ldots, \theta_{N_t}^{(k)}]^{\mathrm{T}}$ containing N_t sampling points is constructed within $[\theta_c^{(k)} - R^{(k)}, \theta_c^{(k)} + R^{(k)}]$, and the peak RT response at each sampling point is calculated to form the observation vector $\mathbf{Y}_{(k)} = [y_1^{(k)}, y_2^{(k)}, \ldots, y_{N_t}^{(k)}]^{\mathrm{T}}$. Subsequently, least squares regression is employed to estimate the parameters of quadratic model $\mathbf{A}_{(k)} = [a_2^{(k)}, a_1^{(k)}, a_0^{(k)}]^{\mathrm{T}}$, where $a_2^{(k)}$, $a_1^{(k)}$, and $a_0^{(k)}$ represent the quadratic coefficient, linear coefficient, and constant term, respectively. This problem is equivalent to the following optimization:

$$\begin{aligned}\hat{\mathbf{A}}_{(k)} &= \arg\min_{\mathbf{A}_{(k)}} \left\|\mathbf{H}_{(k)}\mathbf{A}_{(k)} - \mathbf{Y}_{(k)}\right\|^2 \\ &= \arg\min_{\mathbf{A}_{(k)}} \left(\mathbf{H}_{(k)}\mathbf{A}_{(k)} - \mathbf{Y}_{(k)}\right)^{\mathrm{T}} \left(\mathbf{H}_{(k)}\mathbf{A}_{(k)} - \mathbf{Y}_{(k)}\right)\end{aligned} \tag{24}$$

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where $\mathbf{H}_{(k)}$ is an $N_t \times 3$ Vandermonde matrix

$$\mathbf{H}_{(k)} = \begin{bmatrix} \left(\theta_1^{(k)}\right)^2 & \theta_1^{(k)} & 1 \\ \vdots & \vdots & \vdots \\ \left(\theta_{N_t}^{(k)}\right)^2 & \theta_{N_t}^{(k)} & 1 \end{bmatrix}. \tag{25}$$

The least squares solution of the parameters $\mathbf{A}_{(k)}$ is given by

$$\hat{\mathbf{A}}_{(k)} = \left(\mathbf{H}_{(k)}^{\mathrm{T}}\mathbf{H}_{(k)}\right)^{-1}\mathbf{H}_{(k)}^{\mathrm{T}}\mathbf{Y}_{(k)}. \tag{26}$$

After obtaining $\mathbf{A}_{(k)}$, the vertex position of the quadratic function is derived as

$$\hat{\theta}_{(k)} = -\frac{\hat{a}_1^{(k)}}{2\hat{a}_2^{(k)}}. \tag{27}$$

Subsequently, the search center $\theta_c^{(k+1)}$ is updated to the vertex $\hat{\theta}_{(k)}$ estimated in the current iteration. Following this, a scale-contraction factor γ is introduced to shrink the search radius, i.e., $R^{(k+1)} = \max\{\gamma R^{(k)}, \varepsilon_p\}$, where γ is the golden ratio and ε_p is the preset tolerance. This scale-contraction mechanism enables the method to capture the global trend using a large-scale window in the early iterations for fast guidance, while focusing on a local neighborhood for refined fitting in the later stage. This process is repeated until the update step of the search center is smaller than the preset tolerance ε_p, at which point the final estimated angle $\hat{\theta}_{opt}$ is then taken as the global optimum.

The corresponding range velocity $\hat{v}_r$ is further calculated from $\hat{\theta}_{opt}$. Let the azimuth time span and the fast time span associated with range walk be denoted by $\Delta\eta$ and $\Delta\tau$, respectively. Then, the relationship between $\Delta\eta$ and $\Delta\tau$ is given by

$$\Delta\tau = \frac{2v_r\Delta\eta}{c}. \tag{28}$$

The number of azimuth cells corresponding to $\Delta\eta$ is $L_a = \Delta\eta \times PRF$, and the number of range cells corresponding to $\Delta\tau$ is $L_r = \Delta\tau \times f_s$, where PRF is the pulse repetition frequency and f_s is the sampling frequency. According to the geometric relationship in Fig. 2(a), it follows that:

$$\tan\theta = \frac{L_r}{L_a}. \tag{29}$$

Substituting the expressions of L_a and L_r along with (28) into (29) yields the relationship between $\hat{v}_r$ and $\hat{\theta}_{opt}$ as

$$\hat{v}_r = \frac{c \times PRF \times \tan\hat{\theta}_{opt}}{2f_s}. \tag{30}$$

Based on the estimated range velocity $\hat{v}_r$, the matched filter is constructed as

$$H_{rwc}(f_\tau, \eta) = \exp\left(j\frac{4\pi}{c}f_\tau\hat{v}_r\eta\right). \tag{31}$$

Range walk correction is achieved by performing range FT on $s_{rcc}(\tau, \eta)$ and multiplying it by $H_{rwc}(f_\tau, \eta)$, followed by range IFT. The time domain signal after range walk correction is given by

$$s_{rwc}(\tau, \eta) \approx G_{rc}\sigma_0 \mathrm{sinc}\left(B\left(\tau - \frac{2R_0}{c}\right)\right)\exp\left(-j\frac{4\pi}{\lambda}R(\eta)\right). \tag{32}$$

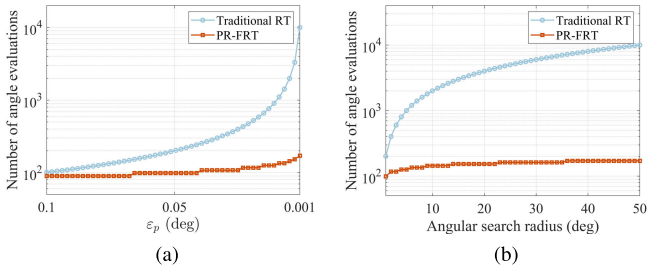


Fig. 3. Number of angle evaluations versus search parameters. (a) Versus search precision. (b) Versus search radius.

It can be observed that, after range walk correction, the range-compressed signal is aligned to the same range cell, thereby achieving effective correction before detection and providing the necessary foundation for subsequent target detection in the range-compressed domain.

B. Computational Complexity Analysis

Let M denote the number of pulses. In the traditional RT, the number of angle search points and range search points are N_θ and N_ρ, respectively. Therefore, the dominant computational complexity of the traditional RT is

$$O(RT) \approx N_\rho N_\theta (M-1). \tag{33}$$

For PR-FRT, let the number of iterations be K_t. The corresponding dominant computational complexity is

$$O(PR-FRT) \approx N_\rho (K_t N_t)(M-1). \tag{34}$$

According to the above derivation, the maximum number of iterations is $K_{\max} = 1 + \lceil(\ln(\varepsilon_p) - \ln(R^{(1)}))/\ln(\gamma)\rceil$, where $\lceil\cdot\rceil$ denotes the ceiling operator.

From (33) and (34), it is evident that under the same M and N_ρ, the complexity difference between the traditional RT and PR-FRT is primarily determined by the number of angle evaluations, i.e., N_θ and $(K_t N_t)$. To quantify this difference, the angle search interval of the traditional RT is set to ε_p, and N_t is set to 9. In this case, the variations of N_θ and $(K_{\max} N_t)$ with respect to the search precision and the search radius are illustrated in Fig. 3. As the search precision increases or the search radius expands, the number of angle evaluations required by the traditional RT increases sharply. In contrast, the number of angle evaluations of PR-FRT grows much more slowly, demonstrating that the proposed PR-FRT method significantly reduces the computational complexity.

IV. EDT-CFAR Detector

In this section, an EDT-CFAR detector operating in the range-compressed domain is proposed for maritime scenes. The detector constructs a double threshold detection mechanism under the GΓD sea clutter model. Specifically, the first threshold is analytically derived from the estimated GΓD parameters and performs cell-by-cell decisions. The second threshold is applied to the cells that pass the first threshold and is adaptively designed to suppress false alarms induced by strong clutter. With this decision mechanism,

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the proposed detector significantly enhances detection performance for weak maritime targets while maintaining the CFAR property.

A. First Threshold Design

GΓD can characterize the statistical properties of sea clutter under different sea states more accurately than traditional statistical models, owing to its high degree of freedom in statistical description [32]. Consequently, it has been widely adopted for the statistical modeling of SAR sea clutter [33], [34], [35], [36].

The probability density function (pdf) of GΓD [33] is given by

$$p(x) = \frac{|\nu|\kappa^{\kappa}}{\sigma\Gamma(\kappa)}\left(\frac{x}{\sigma}\right)^{\kappa\nu-1}\exp\left(-\kappa\left(\frac{x}{\sigma}\right)^{\nu}\right), \quad |\nu|, \kappa, \sigma, x > 0 \tag{35}$$

where x denotes the sea clutter amplitude, and ν, κ, and σ are the power, shape, and scale parameters, respectively. $\Gamma(\cdot)$ denotes the Gamma function. This distribution is relatively versatile, as many classical distributions used in SAR sea clutter statistical modeling can be viewed as special cases under specific parameter constraints. Examples include the Rayleigh ($\nu = 2$ and $\kappa = 1$), Exponential ($\nu = 1$ and $\kappa = 1$), Gamma ($\nu = 1$), Weibull ($\kappa = 1$), and inverse Gamma ($\nu = -1$) distributions.

To achieve accurate parameter estimation, this article adopts a closed-form estimator based on the method-of-log-cumulants (MoLCs) [33]. By estimating the parameters ν, κ, and σ of GΓD based on the sea clutter observation set $\{x_i\}, i = 1, \ldots, N_s$, the analytical approximations for the parameters are obtained as follows:

$$\hat{\kappa} = -\frac{3\hat{c}_3^2 - 2\hat{c}_2^3}{6\hat{c}_3^2} + \sqrt[3]{-\frac{q}{2} + \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}} \tag{36}$$

$$\hat{\nu} = \text{sign}(-\hat{c}_3)\sqrt{\frac{\Phi_o(1, \hat{\kappa})}{\hat{c}_2}} \tag{37}$$

$$\hat{\sigma} = \exp\left(\hat{c}_1 - \frac{\Phi_0(\hat{\kappa}) - \ln(\hat{\kappa})}{\hat{\nu}}\right) \tag{38}$$

where sign$(\cdot)$ denotes the sign function, and the expressions for p and q in (36) are given by

$$p = -\frac{\hat{c}_2^3\left(\hat{c}_2^3 + 3\hat{c}_3^2\right)}{3\hat{c}_3^4} \tag{39}$$

$$q = -\frac{\hat{c}_2^3\left(8\hat{c}_2^6 + 36\hat{c}_2^3\hat{c}_3^2 + 27\hat{c}_3^4\right)}{108\hat{c}_3^6} \tag{40}$$

where $\hat{c}_1$, $\hat{c}_2$, and $\hat{c}_3$ are the sample log-cumulants

$$\hat{c}_1 = \frac{1}{N_s}\sum_{i=1}^{N_s}\ln(x_i) \tag{41}$$

$$\hat{c}_2 = \frac{1}{N_s}\sum_{i=1}^{N_s}(\ln(x_i) - \hat{c}_1)^2 \tag{42}$$

$$\hat{c}_3 = \frac{1}{N_s}\sum_{i=1}^{N_s}(\ln(x_i) - \hat{c}_1)^3. \tag{43}$$

To ensure the existence of a positive real solution in (36), the condition $3\hat{c}_3^2 \le 8\hat{c}_2^3$ is required, which has been reported to be satisfied for most SAR data [33]. For cases where this condition is not satisfied, the parameter estimation scheme reported in [34] can be adopted as an alternative.

By further integrating the pdf, the cumulative distribution function (cdf) can be obtained as

$$F(x) = \int_0^x p(t)\mathrm{d}t = \begin{cases} P\left(\kappa\left(\frac{x}{\sigma}\right)^{\nu}, \kappa\right), & \nu > 0 \\ 1 - P\left(\kappa\left(\frac{x}{\sigma}\right)^{\nu}, \kappa\right), & \nu < 0 \end{cases} \tag{44}$$

where $P(x, a)$ is the regularized incomplete Gamma function [37], defined as $P(x, a) = (1/\Gamma(a))\int_0^x t^{a-1}\mathrm{e}^{-t}dt$, $a, x > 0$.

To achieve effective detection of maritime scene targets in the range-compressed domain, 2-D sliding window detection is performed on the range-compressed data $s_{rwc}(\tau, \eta)$ after CBD processing. The azimuth window length and range window length are set to L and K, respectively. The range-compressed data within the detection window are expressed as

$$\mathbf{s} = \begin{bmatrix} s_{1,1} & \cdots & s_{1,K} \\ \vdots & \ddots & \vdots \\ s_{L,1} & \cdots & s_{L,K} \end{bmatrix}. \tag{45}$$

The first threshold decision aims to screen potential target cells. The amplitude of each cell $s_{l,k}(l = 1, \ldots, L; k = 1, \ldots, K)$ in $\mathbf{s}$ is utilized as the first threshold test statistic $\Lambda_{1k}^{(l)}$, which is then compared with the first threshold T_1. To satisfy the preset first-threshold false alarm probability P_{fa1}, under H_0 hypothesis (only noise), the first threshold T_1 should satisfy the following constraint:

$$P_{fa1} = P\left(\Lambda_{1k}^{(l)} > T_1; H_0\right) = \int_{T_1}^{+\infty} p(t)dt = 1 - F(T_1). \tag{46}$$

By substituting the cdf of GΓD in (44) into (46), the analytical expression of T_1 is obtained as

$$T_1 = \begin{cases} \hat{\sigma}\left(\frac{1}{\hat{\kappa}}P^{-1}\left(1 - P_{fa1}, \hat{\kappa}\right)\right)^{\frac{1}{\hat{\nu}}}, & \hat{\nu} > 0 \\ \hat{\sigma}\left(\frac{1}{\hat{\kappa}}P^{-1}\left(P_{fa1}, \hat{\kappa}\right)\right)^{\frac{1}{\hat{\nu}}}, & \hat{\nu} < 0 \end{cases} \tag{47}$$

where $P^{-1}(x, a)$ denotes the inverse function of $P(x, a)$.

B. Second Threshold Design

The second threshold decision aims to further suppress false alarms by exploiting the energy distribution characteristics of targets in the range-compressed domain. For the kth range cell, the corresponding second threshold test statistic Λ_{2k} is defined as

$$\Lambda_{2k} = \sum_{l=1}^{L}\hat{\kappa}\left(\frac{\Lambda_{1k}^{(l)}}{\hat{\sigma}}\right)^{\hat{\nu}} I\left(\Lambda_{1k}^{(l)} - T_1\right), \quad k = 1, 2, \ldots, K \tag{48}$$

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where $I(\cdot)$ denotes the indicator function defined as

$$I\left(\Lambda_{1k}^{(l)} - T_1\right) = \begin{cases} 1, & \Lambda_{1k}^{(l)} > T_1 \\ 0, & \Lambda_{1k}^{(l)} \le T_1 \end{cases} \tag{49}$$

which is used to select the cells that pass the first threshold. Accordingly, the count of cells that pass the first threshold within each range cell, denoted by N_k, is expressed as

$$N_k = \sum_{l=1}^{L} I\left(\Lambda_{1k}^{(l)} - T_1\right), \quad k = 1, 2, \ldots, K. \tag{50}$$

Under H_0 hypothesis, the count N_k follows a binomial distribution. Its probability mass function (PMF) is given by

$$P\left(N_k; H_0\right) = C_L^{N_k} \left(P_{fa1}\right)^{N_k} \left(1 - P_{fa1}\right)^{L - N_k} \tag{51}$$

where $C_L^{N_k}$ denotes the binomial coefficient. Accordingly, the expectation of N_k is

$$E\left(N_k; H_0\right) = LP_{fa1}. \tag{52}$$

For different values of N_k, the conditional statistical characteristics of Λ_{2k} vary accordingly. To ensure the CFAR property of the detector, the second threshold should be designed as an adaptive function of N_k, denoted by $T_2(N_k)$. Accordingly, the second-threshold false alarm probability P_{fa2} is expressed as

$$P_{fa2} = \begin{cases} \sum_{N_k=0}^{L} P(\Lambda_{2k} > T_2(N_k); H_0)\ P(N_k; H_0), & \hat{\nu} > 0 \\ \sum_{N_k=0}^{L} P(\Lambda_{2k} < T_2(N_k); H_0)\ P(N_k; H_0), & \hat{\nu} < 0. \end{cases} \tag{53}$$

Since the second threshold will not be triggered when $N_k = 0$, (53) can be further simplified as

$$P_{fa2} = \begin{cases} \sum_{N_k=1}^{L} P\left(\tilde{\Lambda}_{2k} > \tilde{T}_2(N_k); H_0\right) P\left(N_k; H_0\right), & \hat{\nu} > 0 \\ \sum_{N_k=1}^{L} P\left(\tilde{\Lambda}_{2k} < \tilde{T}_2(N_k); H_0\right) P\left(N_k; H_0\right), & \hat{\nu} < 0 \end{cases} \tag{54}$$

where

$$\tilde{\Lambda}_{2k} = \Lambda_{2k} - N_k \hat{\kappa} \left(\frac{T_1}{\hat{\sigma}}\right)^{\hat{\nu}} \tag{55}$$

$$\tilde{T}_2\left(N_k\right) = T_2\left(N_k\right) - N_k \hat{\kappa} \left(\frac{T_1}{\hat{\sigma}}\right)^{\hat{\nu}}. \tag{56}$$

The window-level false alarm probability P_{fa} is defined as the probability that at least one false alarm occurs within the K range cells corresponding to the detection window. The relationship between P_{fa} and P_{fa2} is as follows:

$$P_{fa} = 1 - \left(1 - P_{fa2}\right)^K. \tag{57}$$

Following the design principle of a constant overall false alarm probability P_{fa}, P_{fa2} should be set as:

$$P_{fa2} = 1 - \left(1 - P_{fa}\right)^{\frac{1}{K}}. \tag{58}$$

In scenarios where the target scattering density is unavailable, this article follows the approach presented in [29] to derive the adaptive mapping relationship between $\tilde{T}_2(N_k)$ and N_k for each range cell:

$$\frac{P_{fa2}}{1 - \left(1 - P_{fa1}\right)^L} = \begin{cases} P\left(\tilde{\Lambda}_{2k} > \tilde{T}_2(N_k); H_0\right), & \hat{\nu} > 0 \\ P\left(\tilde{\Lambda}_{2k} < \tilde{T}_2(N_k); H_0\right), & \hat{\nu} < 0. \end{cases} \tag{59}$$

To determine the second threshold $T_2(N_k)$, it is necessary to solve for the threshold corresponding to the tail probability in (59). This article introduces the Lugannani–Rice (L–R) saddle point approximation method [38] to obtain an accurate approximation of the tail probability, thereby determining the corresponding threshold. This method is based on the saddle point analysis of the cumulant generating function (CGF). Compared with the Gaussian approximation and moment-matching Gamma approximation, it exhibits higher accuracy in the extreme tail probability region, making it well-suited for threshold derivation under the low false alarm probability addressed in this section.

First, under the H_0 hypothesis, $\tilde{\Lambda}_{2k}$ can be expressed as the sum of independent and identically distributed random variables, given by

$$\begin{aligned} \tilde{\Lambda}_{2k} &= \sum_{j=1}^{N_k} \left(\hat{\kappa} \left(\frac{\Lambda_{1k}^{(j)}}{\hat{\sigma}} \right)^{\hat{\nu}} - \hat{\kappa} \left(\frac{T_1}{\hat{\sigma}} \right)^{\hat{\nu}} \right) \\ &= \sum_{j=1}^{N_k} \left(U_j - u_1\right) = \sum_{j=1}^{N_k} V_j \end{aligned} \tag{60}$$

where $V_j (j = 1, \ldots, N_k)$ denotes the corresponding random variable in the Gamma domain.

Since the derivations for the cases $\hat{\nu} > 0$ and $\hat{\nu} < 0$ are essentially identical, the following derivation focuses on the more common case $\hat{\nu} > 0$ in practical applications.

Let V denote the representative random variable with the same distribution as V_j. Then, under H_0 hypothesis, the moment generating function (MGF) of V can be expressed as

$$M_V(s) = \exp\left(-su_1\right)(1 - s)^{-\hat{\kappa}} \frac{Q\left((1 - s)u_1, \hat{\kappa}\right)}{Q\left(u_1, \hat{\kappa}\right)}, \quad s < 1 \tag{61}$$

where $Q(x, a)$ denotes the regularized incomplete Gamma function, defined as $Q(x, a) = (1/\Gamma(a)) \int_x^\infty t^{a-1} \mathrm{e}^{-t} dt,\ a, x > 0$.

Accordingly, the CGF of V, $K_V(s) = \ln(M_V(s))$, and its first and second derivatives are given by

$$\begin{aligned} K_V(s) = &-su_1 - \hat{\kappa} \ln(1 - s) + \ln(Q((1 - s)u_1, \hat{\kappa})) \\ &- \ln(Q(u_1, \hat{\kappa})) \end{aligned} \tag{62}$$

$$K_V'(s) = -u_1 + \frac{\hat{\kappa}}{1 - s} + u_1 r\left((1 - s)u_1\right) \tag{63}$$

$$K_V''(s) = \frac{\hat{\kappa}}{(1 - s)^2} - {u_1}^2 r'\left((1 - s)u_1\right) \tag{64}$$

where $r(x)$ and $r'(x)$ are defined in (65) and (66), and $\Gamma(x, \hat{\kappa})$ denotes the incomplete Gamma function

$$r(x) = \frac{x^{\hat{\kappa}-1} e^{-x}}{\Gamma(x, \hat{\kappa})} \tag{65}$$

$$r'(x) = r(x) \left(\frac{\hat{\kappa} - 1}{x} - 1 + r(x) \right). \tag{66}$$

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From (60), $\tilde{\Lambda}_{2k}$ is the sum of N_k variables V_j. Accordingly, the CGF of $\tilde{\Lambda}_{2k}$ can be obtained as

$$K_{\tilde{\Lambda}_{2k}}(s) = N_k K_V(s) \tag{67}$$

$$K_{\tilde{\Lambda}_{2k}}'(s) = N_k K_V'(s) \tag{68}$$

$$K_{\tilde{\Lambda}_{2k}}''(s) = N_k K_V''(s). \tag{69}$$

For a given candidate threshold T_n, the corresponding saddle point $\hat{s}_n$ is obtained by solving the saddle point equation $K_{\tilde{\Lambda}_{2k}}'(s) = T_n$, upon which the following variables are introduced:

$$w_n = \text{sign}(\hat{s}_n)\sqrt{2\left(\hat{s}_n T_n - K_{\tilde{\Lambda}_{2k}}(\hat{s}_n)\right)} \tag{70}$$

$$u_n = \hat{s}_n \sqrt{K_{\tilde{\Lambda}_{2k}}''(\hat{s}_n)}. \tag{71}$$

According to the L–R tail approximation [39], the right tail probability of $\tilde{\Lambda}_{2k}$ under H_0 hypothesis can be expressed as

$$P\left(\tilde{\Lambda}_{2k} > T_n; H_0\right) \approx \Phi(-w_n) + \phi(w_n)\left(\frac{1}{u_n} - \frac{1}{w_n}\right) \tag{72}$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ denote the pdf and cdf of the standard Gaussian distribution, respectively.

For each possible value of N_k, $\tilde{T}_2(N_k)$ is numerically solved such that the conditional right tail probability of $\tilde{\Lambda}_{2k}$ under H_0 hypothesis, approximated by (72), equals the prescribed tail probability specified in (59). According to the relationship in (56), the second threshold $T_2(N_k)$ is then obtained as follows:

$$T_2(N_k) = \tilde{T}_2(N_k) + N_k \hat{\kappa}\left(\frac{T_1}{\hat{\sigma}}\right)^{\hat{\nu}}. \tag{73}$$

In summary, this article proposes an integrated detection method that cascades CBD with the EDT-CFAR detector. This method first employs the PR-FRT-based CBD strategy to effectively correct the range migration, and then designs the EDT-CFAR detector to achieve CFAR detection. The flowchart is illustrated in Fig. 4.

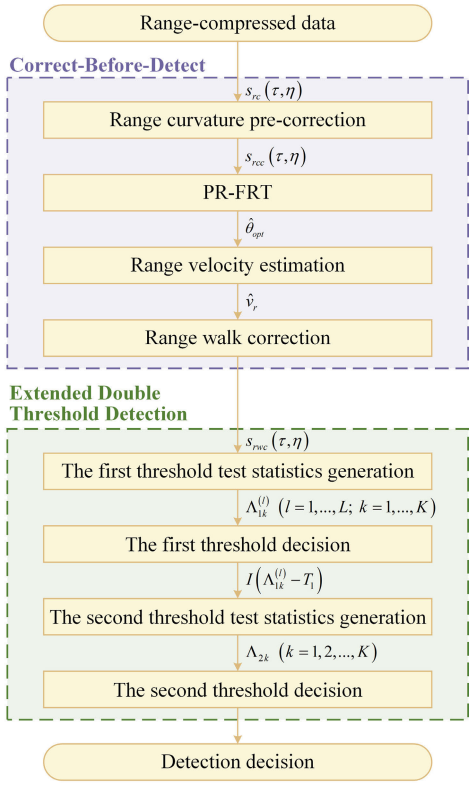


Fig. 4. Flowchart of maritime scene target detection in the range-compressed domain.

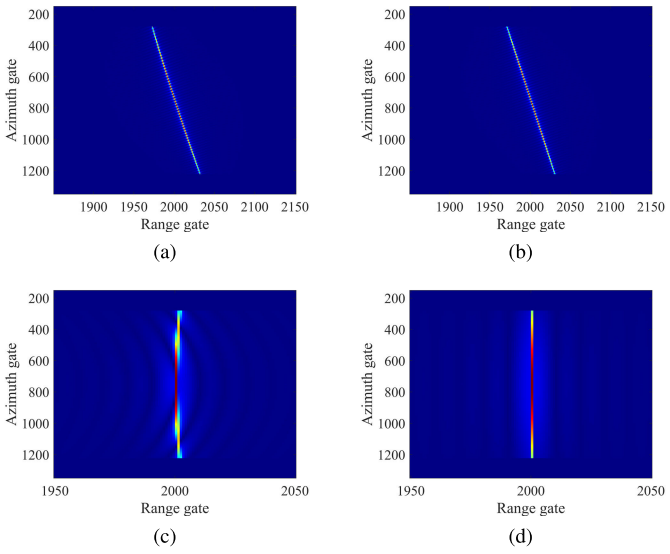


Fig. 5. Comparison of range-compressed signals under different CBD correction steps. (a) Original signal. (b) After range curvature precorrection only. (c) After range walk correction only. (d) After range curvature precorrection and range walk correction.

V. Experimental Validation

A. Simulation Analysis

1) Performance Analysis of the CBD Method: This section validates the effectiveness of the proposed CBD method through simulation analysis. The SAR simulation parameters are listed in Table I, and the operating mode is the stripmap mode. In the simulation, the point target has a range velocity of 200 m/s and an azimuth velocity of 100 m/s. Fig. 5 illustrates the range-compressed signal results at different correction steps. Specifically, Fig. 5(a) shows the signal after range compression, where significant range migration is observed. To analyze each procedure in the CBD method, Fig. 5(b) and (c) presents the results obtained by applying only range curvature precorrection and only range walk correction, respectively. Fig. 5(d) shows the result processed by the complete flow of the proposed CBD method. As can be seen, the proposed method effectively corrects the range migration induced by target and platform motion, thereby aligning the envelope of the range-compressed signal to the same range cell.

To further evaluate the range velocity estimation performance of the proposed PR-FRT method under different SNRs, the background clutter component following GΓD is added to the above signal, with the parameters set as $\nu = 2.24$, $\kappa = 0.85$, and $\sigma = 13.61$. The SNR is varied from −15 to 5 dB in 1-dB steps by adjusting the relative amplitudes of the signal and the background clutter component. Subsequently, the traditional RT and the proposed PR-FRT are employed for range velocity estimation. The angle search range for both methods is set to

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TABLE I
SAR SIMULATION PARAMETERS

Parameter	Value	Unit
Carrier frequency	5.40	GHz
Signal bandwidth	60	MHz
Sampling frequency	66.67	MHz
Pulse duration	45	μs
Slant range of scene center	900	km
Altitude	755	km
Effective radar velocity	7569.77	m/s
Pulse repetition frequency	1410.92	Hz
Squint angle	0	deg
Antenna length	10	m

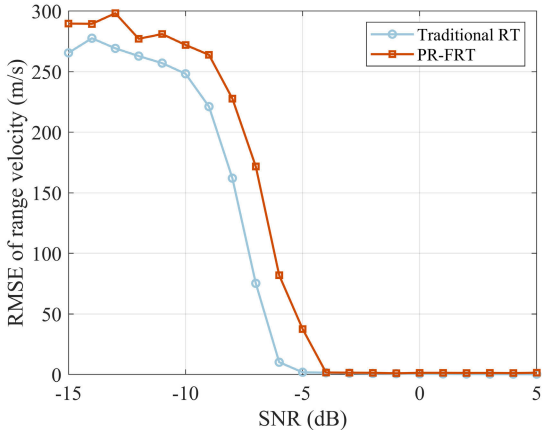


Fig. 6. RMSE of range velocity estimation versus SNR.

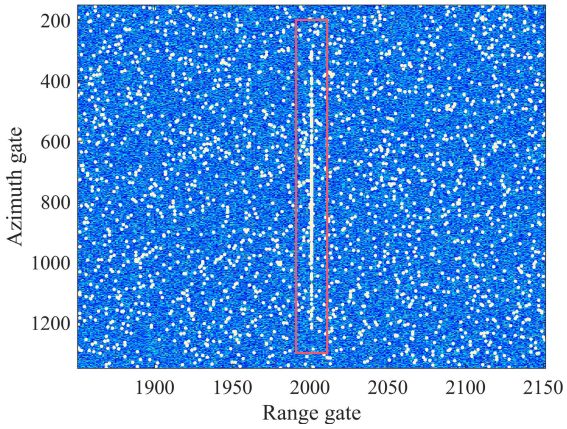


Fig. 7. Decisions of the first threshold.

$[-5°, 5°]$, and the angle search interval of the traditional RT is set to 0.01°. For PR-FRT, the number of sampling points N_t is set to 9, and the preset tolerance ε_p is set to 0.01°. For each SNR, 1000 independent Monte Carlo trials are conducted, and the root-mean-square error (RMSE) is adopted as the performance metric. The corresponding estimation results are illustrated in Fig. 6.

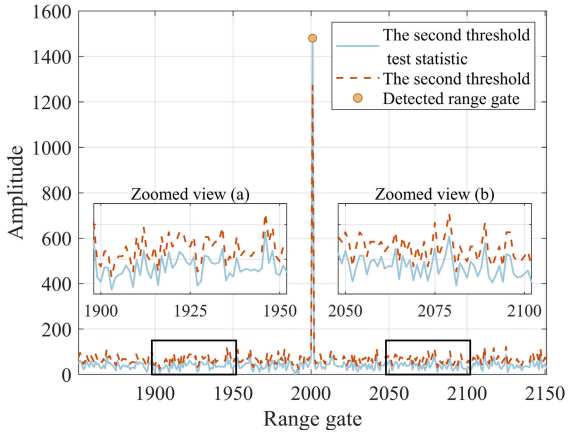


Fig. 8. Decisions of the second threshold. (a) Zoomed view around range gates 1900-1950. (b) Zoomed view around range gates 2050-2100.

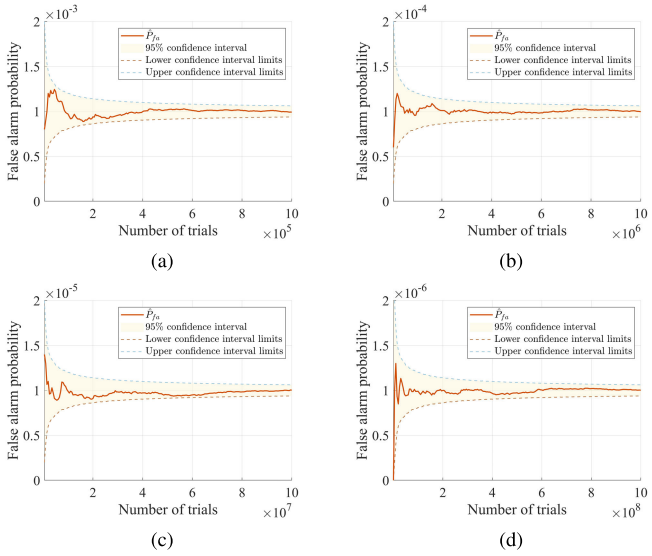


Fig. 9. False alarm probability versus the number of trials. (a) $P_{fa} = 1\times10^{-3}$. (b) $P_{fa} = 1 \times 10^{-4}$. (c) $P_{fa} = 1 \times 10^{-5}$. (d) $P_{fa} = 1 \times 10^{-6}$.

Simulation results indicate that, to achieve accurate range velocity estimation, the minimum required SNR for the traditional RT is approximately −5 dB, while that for PR-FRT is approximately −4 dB, representing a performance degradation of about 1 dB compared with the traditional RT. For the subsequent EDT-CFAR detector, a detection probability of 90% is achieved at an SNR of approximately 1.5 dB with a false alarm probability of 1×10^{-6}. Therefore, within the effective SNR range satisfying the detection requirements, the proposed PR-FRT provides reliable range velocity estimation. Furthermore, the corresponding methods are implemented on a computer equipped with an Intel i7-12700 processor. The average runtime of the traditional RT is 0.4038 s, whereas that of PR-FRT is 0.0417 s. This demonstrates that the proposed PR-FRT significantly enhances computational efficiency, thereby enabling efficient CBD processing.

2) Performance Analysis of the EDT-CFAR Detector: This section evaluates the detection performance of the proposed EDT-CFAR detector through simulation analysis. Background

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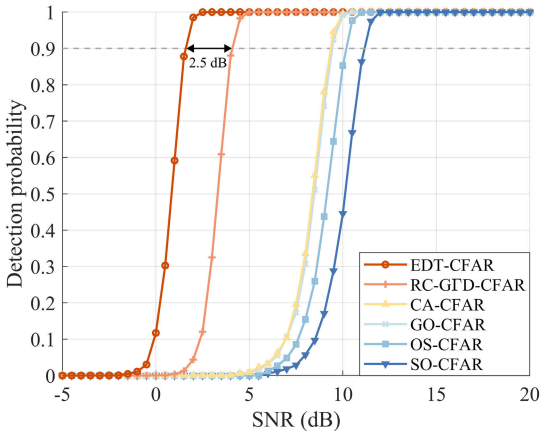


Fig. 10. Detection probability versus SNR.

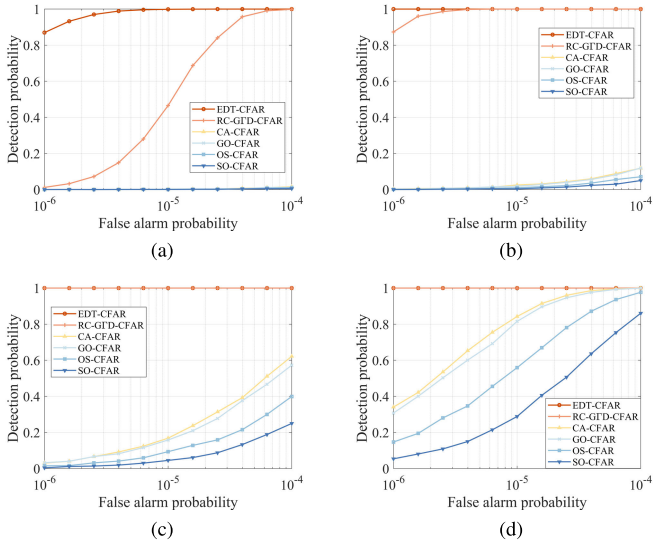


Fig. 11. ROC curves under different SNRs. (a) SNR = 1.5 dB. (b) SNR = 4 dB. (c) SNR = 6 dB. (d) SNR = 8 dB.

clutter following the GΓD model is added to the signal, with the parameters kept consistent with those previously mentioned, and the SNR in the range-compressed domain is set to 3 dB. In the detection procedure, the proposed CBD method is first applied to correct range migration. On this basis, target detection is performed using the EDT-CFAR detector, with the false alarm probability set to $P_{fa} = 1 \times 10^{-6}$.

The first-threshold false alarm probability is set to $P_{fa1} = 5 \times 10^{-3}$, guided by (52), to preserve target cells while partially suppressing background clutter. The first threshold decisions are illustrated in Fig. 7, where the white points denote the candidate cells that have passed the first threshold. It is observed that, although the first threshold suppresses low-intensity background clutter, a considerable number of false alarms remain. The red box marks the target signal location. As observed, the target signal is substantially preserved after the first threshold, thereby providing a reliable set of candidate cells for the second threshold decision.

Building upon this, the test statistics for the second threshold are constructed using the cells that pass the first threshold. The adaptive threshold is then established to effectively suppress false alarms induced by strong clutter under the CFAR constraint. The second threshold decisions are illustrated in Fig. 8. Simulation results demonstrate that the proposed detector successfully detects the target. As observed from the zoomed views, the test statistics in clutter-only regions remain below the adaptive second threshold, enabling reliable detection of weak targets in maritime scenes.

To verify the CFAR property of the proposed detector, Monte Carlo simulations are conducted under clutter-only conditions to statistically estimate the false alarm probability. The clutter parameters are kept consistent with those previously described. For a given false alarm probability P_{fa}, $1000/P_{fa}$ independent trials are performed, and the empirical false alarm probability $\hat{P}_{fa}$ is defined as the ratio of the number of false alarms to the number of trials. Fig. 9 illustrates the evolution of $\hat{P}_{fa}$ versus the number of trials for P_{fa} values of 1×10^{-3}, 1×10^{-4}, 1×10^{-5}, and 1×10^{-6}, together with the 95% confidence interval of P_{fa} relative to the number of trials. It is observed that $\hat{P}_{fa}$ exhibits fluctuations when the number of trials is small. As the number of trials increases, $\hat{P}_{fa}$ gradually converges to the theoretical P_{fa} and remains consistently within the 95% confidence interval. These results statistically validate that the proposed detector maintains CFAR under different false alarm probability constraints, which is consistent with the theoretical derivations in Section IV.

To evaluate the detection performance of the proposed EDT-CFAR detector under different SNRs, Monte Carlo simulations are conducted with the same simulation setup. The window-level false alarm probability is set to $P_{fa} = 1 \times 10^{-6}$, and the SNR in the range-compressed domain is varied from -5 to 20 dB with a step size of 0.5 dB. For each SNR, 5000 independent trials are performed. The detectors used for comparison include RC-GΓD-CFAR [28], cell-averaging CFAR (CA-CFAR) [40], greatest-of CFAR (GO-CFAR) [41], smallest-of CFAR (SO-CFAR) [42], and ordered-statistics CFAR (OS-CFAR) [43]. For the four classical 1-D CFAR detectors, the number of reference cells is set to 24, and the number of guard cells is set to 8 in total. Fig. 10 illustrates the detection probability versus SNR for different detectors. To achieve a detection probability of 90%, the proposed EDT-CFAR requires an SNR of approximately 1.5 dB, whereas the corresponding SNR for RC-GΓD-CFAR is about 4 dB, indicating an approximately 2.5-dB reduction in the SNR requirement, as annotated in Fig. 10. In addition, the four 1-D CFAR detectors require higher SNRs to reach the same detection probability. These results indicate that the proposed detector effectively enhances the detection performance under low SNR conditions.

To further evaluate the detection performance of the proposed EDT-CFAR detector under different false alarm probability settings and to provide a quantitative comparison with existing detectors, the receiver operating characteristic (ROC) curves are plotted under SNRs of 1.5, 4, 6, and 8 dB, respectively. For each SNR, the false alarm probability varies logarithmically from 1×10^{-6} to 1×10^{-4}, and 5000

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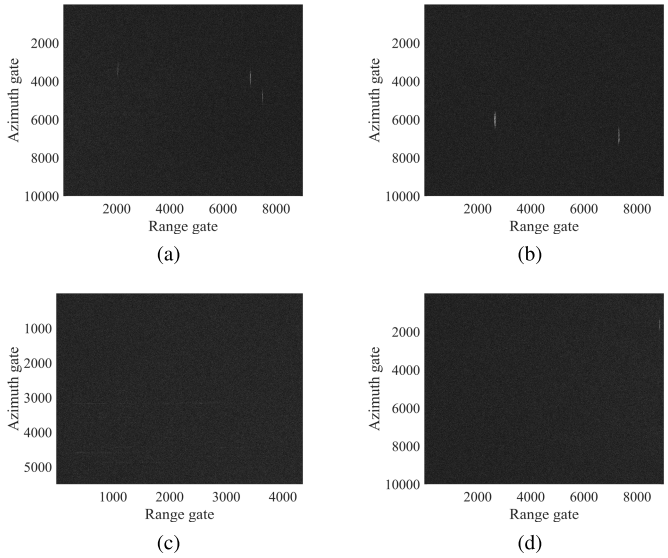


Fig. 12. Range-compressed measured data. (a) Data 1. (b) Data 2. (c) Data 3. (d) Data 4.

independent trials are conducted for each false alarm probability. The other parameter settings are kept the same as those in the preceding experiments. The corresponding ROC curves are shown in Fig. 11. As can be observed, when the SNR is 1.5 dB, EDT-CFAR maintains higher detection probabilities over the false alarm probability range, whereas the detection probability of RC-GΓD-CFAR gradually increases with the false alarm probability, and those of the four 1-D CFAR detectors are close to zero. When the SNR increases to 4 dB, the detection probability of EDT-CFAR remains 1 over the false alarm probability range, while the detection probability of RC-GΓD-CFAR is noticeably improved. Meanwhile, the detection probabilities of the four 1-D CFAR detectors remain relatively low. As the SNR further increases to 6 and 8 dB, the detection probabilities of the four 1-D CFAR detectors gradually increase, but they remain generally lower than those of EDT-CFAR, especially under low false alarm probability. These results demonstrate that the proposed EDT-CFAR detector achieves improved detection performance under different false alarm probability settings.

B. Measured Data Validation

To further validate the effectiveness of the proposed method in practical maritime scenes, four sets of measured data acquired by GF-3 spaceborne SAR, referred to as Data 1, Data 2, Data 3, and Data 4, are employed for target detection in the range-compressed domain, as shown in Fig. 12(a)–(d), respectively. The main parameters of GF-3 are listed in Table II. A 2-D sliding window detection strategy is adopted, where the range window length is set to 500 with a step size of 480, and the azimuth window length is set to 800 with a step size of 400. The first-threshold false alarm probability of the EDT-CFAR detector is set to $P_{fa1} = 5 \times 10^{-3}$, and the overall false alarm probability is set to $P_{fa} = 1 \times 10^{-6}$.

For Data 1, a total of three targets are detected and labeled as Target 1, Target 2, and Target 3. For Data 2, two targets are detected and labeled as Target 1 and Target 2. In each dataset, the targets are labeled in ascending order of range gate index. The target detection windows and imaging results for Data 1 and Data 2 are shown in Fig. 13. For each target, the left and middle columns show the first threshold decisions and second threshold decisions for the corresponding detection window, respectively, while the right column shows the target imaging result. The imaging results show that the five detected targets in Data 1 and Data 2 are all ships. These results demonstrate that the proposed detection method, cascading CBD with the EDT-CFAR detector, achieves effective ship detection in maritime scenes.

TABLE II
SPACEBORNE SAR PARAMETERS

Parameter	Value	Units
Carrier frequency	5.40	GHz
Signal bandwidth	60	MHz
Sampling frequency	66.67	MHz
Pulse duration	45	μs
Near range	860.00	km
Pulse repetition frequency	1410.92	Hz
Squint angle	0	deg

For Data 3, one target is detected with the corresponding results presented in Fig. 14. As shown in Fig. 14(a), the target exhibits significant range migration in the range-compressed domain. The SNR of this target is approximately 3.5 dB, indicating weak signal characteristics. This SNR is defined as

$$\mathrm{SNR} = 10\log_{10}\left(\frac{\bar{P}_{\mathrm{sig}}}{\bar{P}_{\mathrm{noise}}}\right) \tag{74}$$

where $\bar{P}_{\mathrm{sig}}$ denotes the mean power of signal, and $\bar{P}_{\mathrm{noise}}$ denotes the mean power of noise. Fig. 14(b) demonstrates that the CBD method effectively corrects the range migration. Furthermore, the estimated range velocity is −144.06 m/s, which significantly exceeds the typical speed of ships, suggesting that the weak echoes likely correspond to an aerial target over the sea background. Subsequently, as shown in Fig. 14(c) and (d), the double threshold decision successfully suppresses false alarms and achieves target detection. The imaging result in Fig. 14(e) suggests that the target is potentially an aircraft. To further identify the target, the temporal and spatial information of this data is matched with the Automatic Dependent Surveillance-Broadcast (ADS-B) database. The validation results indicate that the target is highly correlated with the flight UAE540, a Boeing 777-300ER (B77W) aircraft, over the northern Arabian Sea during the same period, and an illustration of the B77W aircraft is shown in Fig. 14(f).

For Data 4, two targets are detected, and the processing results are shown in Fig. 15. Specifically, Fig. 15(a)–(c) corresponds to Target 1, while Fig. 15(d)–(f) corresponds to Target 2. As can be observed from Fig. 15(a) and (b), the weak Target 1 is effectively detected through the cascaded decisions of the first and second thresholds. The estimated range velocity

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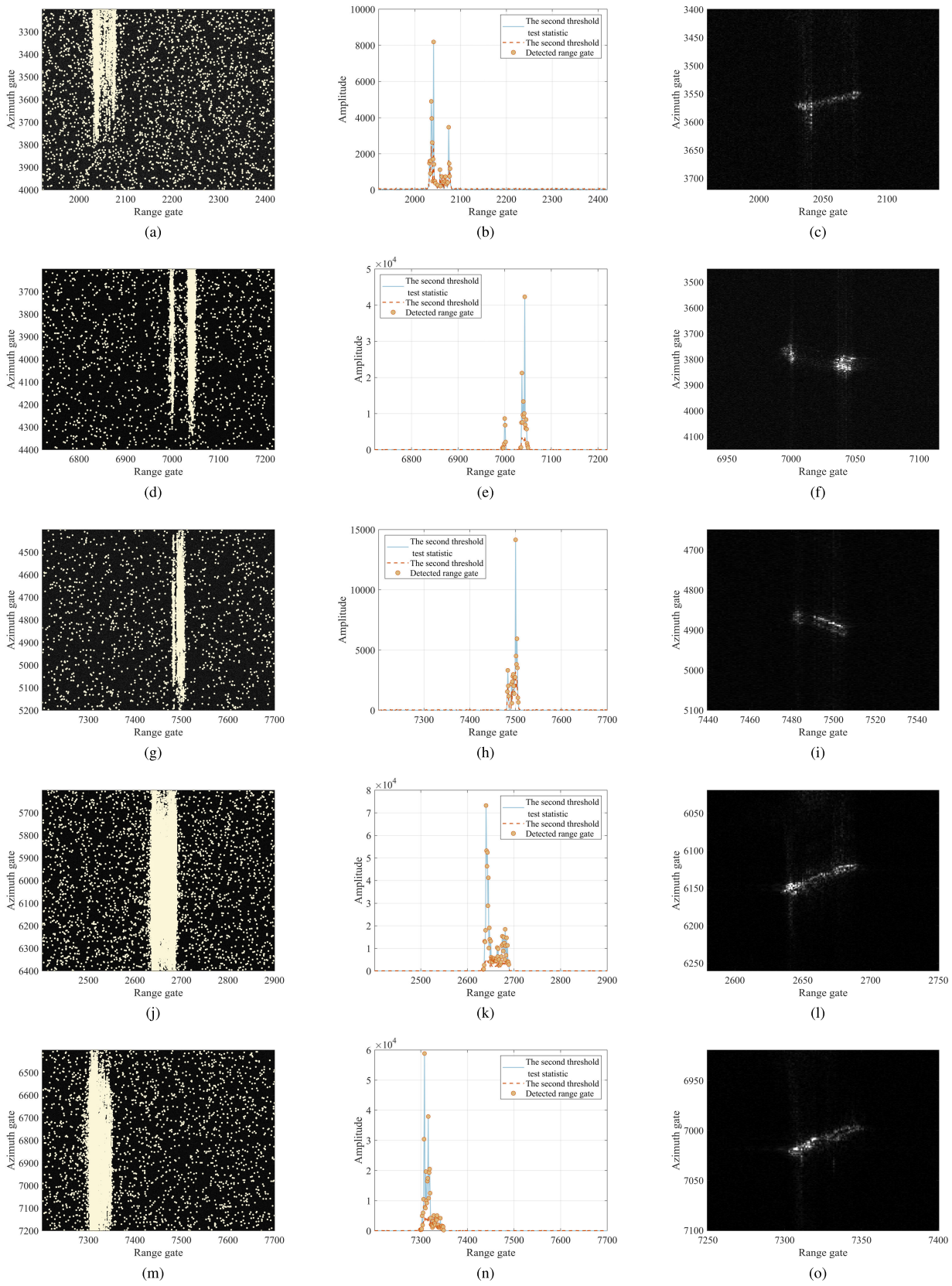


Fig. 13. Processing results for Data 1 and Data 2. For each target, the first threshold decisions, the second threshold decisions, and the imaging result are shown from left to right. (a)–(c) Target 1 in Data 1. (d)–(f) Target 2 in Data 1. (g)–(i) Target 3 in Data 1. (j)–(l) Target 1 in Data 2. (m)–(o) Target 2 in Data 2.

of this target is −144.33 m/s. Similarly, according to the imaging result and the ADS-B database matching result, this target is inferred to be an aircraft, corresponding to flight UAE526 with aircraft type B77W. The imaging result of Target

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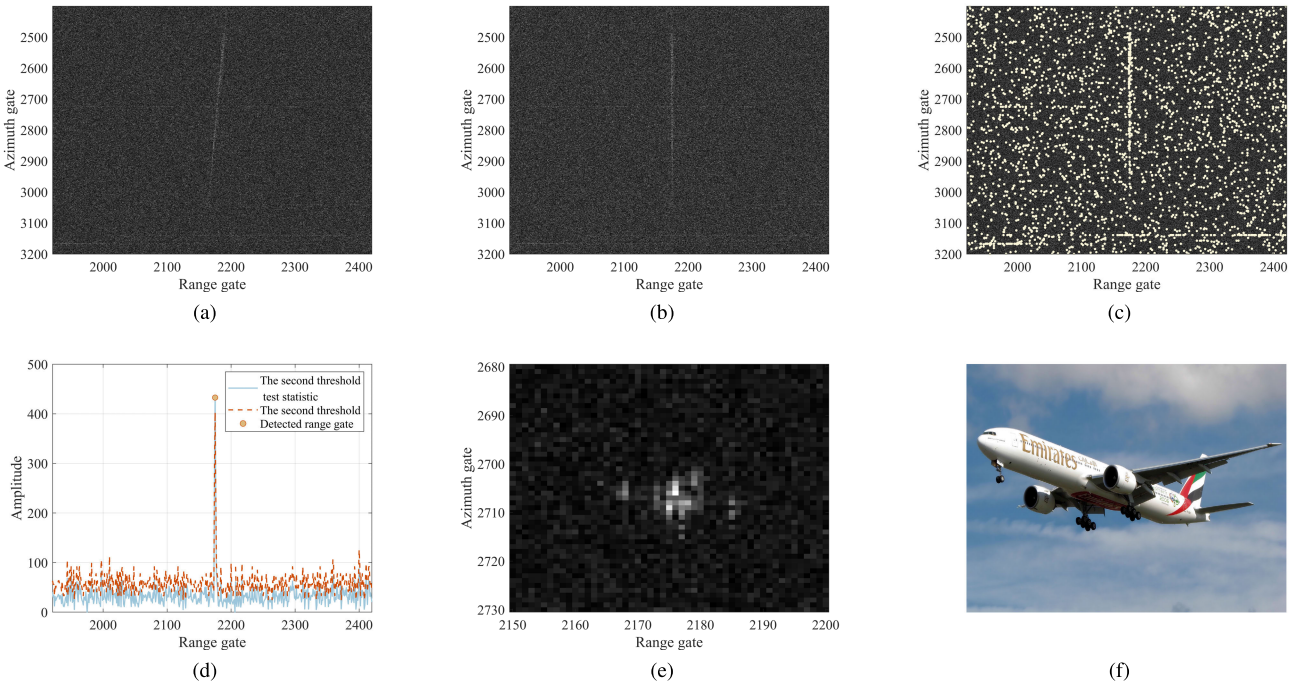


Fig. 14. Processing results for Data 3. (a) Target detection window. (b) After CBD processing. (c) Decisions of the first threshold. (d) Decisions of the second threshold. (e) Imaging result. (f) Illustration of the B77W aircraft.

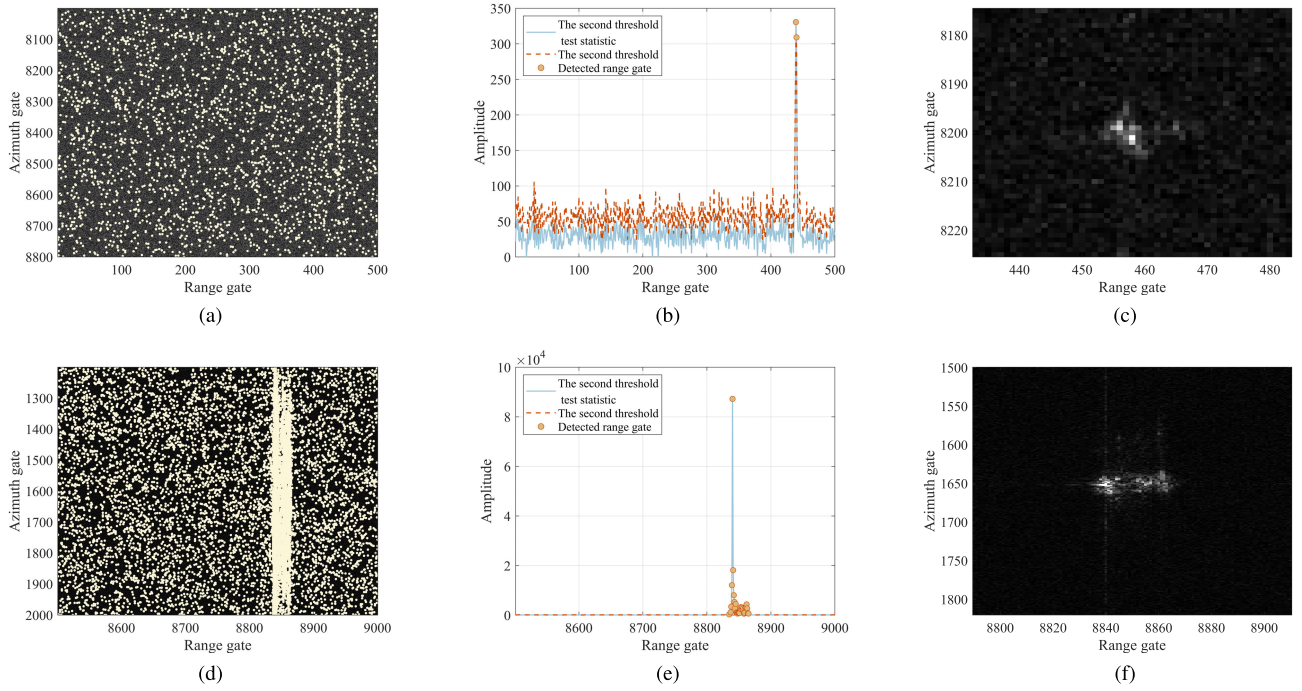


Fig. 15. Processing results for Data 4. For each target, the first threshold decisions, the second threshold decisions, and the imaging result are shown from left to right. (a)–(c) Target 1. (d)–(f) Target 2.

2 suggests that this target is a ship. These results show that Data 4 contains both an aircraft and a ship, further validating the detection capability and applicability of the proposed detection method in maritime scenes where ships and aircraft coexist.

Table III further summarizes the detection results of the comparison detectors and the detailed information of the four measured data, including the sea state, target category, SNR, and estimated target size for each data. As listed in the table, the four measured data contain a total of eight targets, including six ships and two aircraft. The SNRs of these targets range from 3.2 to 10.9 dB, indicating that the measured data cover maritime targets with different echo intensities. CA-CFAR and RC-GTD-CFAR are adopted as comparison detectors. The

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TABLE III
DETAILED INFORMATION AND DETECTION RESULTS FOR MEASURED DATA

Measured Data	Data 1			Data 2		Data 3	Data 4	
Sea state	Sea State 4			Sea State 2		Sea State 4	Sea State 4	
Target ID	Target 1	Target 2	Target 3	Target 1	Target 2	Target 1	Target 1	Target 2
Target category	Ship	Ship	Ship	Ship	Ship	Aircraft	Aircraft	Ship
SNR	4.2 dB	9.3 dB	6.6 dB	10.9 dB	9.3 dB	3.5 dB	3.2 dB	8.0 dB
Target size	Length: 215 m Width: 28 m	Length: 322 m Width: 65 m	Length: 183 m Width: 33 m	Length: 257 m Width: 33 m	Length: 220 m Width: 28 m	Length: 75 m Wingspan: 62 m	Length: 71 m Wingspan: 62 m	Length: 146 m Width: 26 m
CA-CFAR	Not detected	Detected	Not detected	Detected	Not detected	Not detected	Not detected	Detected
RC-GΓD-CFAR	Detected	Detected	Detected	Detected	Detected	Not detected	Not detected	Detected
EDT-CFAR	Detected	Detected	Detected	Detected	Detected	Detected	Detected	Detected

results indicate that CA-CFAR fails to detect some of the ships and both aircraft, while RC-GΓD-CFAR detects all ships but fails to detect both aircraft. By comparison, the proposed EDT-CFAR successfully detects all targets in the four measured datasets.

The experimental results on measured data validate the effectiveness of the proposed integrated CBD and EDT-CFAR detection method for maritime target detection. This method provides robust detection performance for both conventional ships and aerial targets over the sea background, thereby extending the scope of detectable targets for spaceborne SAR maritime target detection.

VI. CONCLUSION

To address the suboptimal detection performance of weak targets in maritime scenes, this article proposes a range-compressed domain detection method that cascades a CBD strategy with an EDT-CFAR detector. First, range curvature is corrected by leveraging prior platform velocity information. Furthermore, PR-FRT is developed to efficiently estimate the range velocity, thereby enabling range walk correction. On this basis, the EDT-CFAR detector is designed under the GΓD sea clutter model, with the double thresholds analytically derived to satisfy the CFAR criterion in the range-compressed domain. Experimental results demonstrate that the proposed integrated method achieves enhanced detection performance under low SNR conditions compared with existing CFAR-based methods. This work provides a feasible technical solution for weak target detection in spaceborne SAR maritime scenes and offers critical guidance for the subsequent imaging.

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